

Automatic Approach Based on Deep Learning for Tea Leaf Disease Detection

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Abstract—Tea production is an essential agricultural business, but it is vulnerable to various illnesses that can reduce productivity and quality. This study uses Convolutional Neural Networks (CNNs) to provide a unique technique for automated categorization of tea leaf illnesses. An extensive collection of 900 high-resolution pictures illustrating eight different types of leaf spot (Red leaf spot, Algal leaf spot, Bird's eyespot, grey blight, White spot, Anthracnose, and Brown spot). Tea leaf diseases (blight.) were curated. A cutting-edge CNN model architecture was created and trained to utilize a deep learning framework and obtained 84.5% classification accuracy. The model has standard criteria such as accuracy, precision, recall, and F1-score were used to assess performance. The outcomes are essential for the tea business since the automated disease classification system can give real-time illness categorization. Tea estates can be monitored in real-time for early disease identification and intervention. This may result in increased tea production, improved crop management, and decreased economic losses for tea producers.

Keywords—Tea Leaf, Disease, Deep Learning, CNN, Image Processing

I. INTRODUCTION

Tea has a particular place in the hearts and cultures of people worldwide. Tea originated in ancient China and is derived from the *Camellia sinensis* plant. Its history dates back centuries. It not only has a lovely perfume and a range of tastes, but it also offers several health advantages. Each kind of Tea, such as black, green, oolong, and white, has a different taste profile and nutritional benefits. Tea is enjoyed for its soothing warmth on a cold morning, its relaxing effects for relaxation, or its function as a social catalyst during the holidays. It continues to be a vital part of our everyday life, signifying togetherness, tradition, and well-being. It is a beverage that crosses boundaries, cultures, and generations, bringing people together over a common interest.

Tea diseases present substantial difficulties to the worldwide tea business, endangering tea estates' health and output. These illnesses include fungal, bacterial, and viral infections that may harm tea plant development, leaf quality, and crop productivity. Algal leaf spot, grey blight, red leaf spot, and brown blight are all prevalent tea illnesses. These

infections may cause tea plant leaf lesions, discoloration, defoliation, and even death. Tea disease control is critical for preserving tea output and quality. To reduce the burden of these illnesses, integrated pest management measures such as good sanitation, frequent monitoring, and the adoption of disease-resistant tea cultivars are critical. Addressing the issues presented by tea diseases is essential to ensure the continuing supply of this popular beverage and the livelihoods of tea producers throughout the globe.

The fungus *Cercospora coffeeicola* causes red leaf spots, often known as red rust, which damages tea plants and other crops. It develops small, scarlet, or brown lesions on the leaves that may grow and cover a significant percentage of the leaf surface. The illness may impair photosynthesis, lower output, and degrade tea quality. It thrives in warm, humid climates, making them more vulnerable to its spread. Effective management includes cultural practices, trimming, proper plant spacing, and fungicide spraying. Illness-resistant tea varieties may also help to mitigate the consequences of the illness. Regular observation and early discovery are critical for averting catastrophic outbreaks and reducing economic losses in tea farming[1].

Algal leaf spot, also known as algal leaf spot disease or green alga leaf spot, is a plant disease predominantly affecting aquatic vegetation caused by green algae. It causes tiny, green to brownish-green blotches or regions on the foliage of infected plants, hindering their ability to photosynthesize. It can occasionally be found on various plants in terrestrial environments, particularly in areas with high humidity and excess moisture. Managing algal leaf spots requires maintaining good water quality in aquatic environments, controlling algal growth with algaecides or physically removing afflicted foliage, improving air circulation, reducing excessive moisture, and practicing good plant sanitation. It is essential to note that algal leaf spot is distinct from fungal leaf spot diseases, which are more prevalent and destructive to various plants [2].

Bird's eye spot, also called Bird's eye rust or Bird's eye disease, is a fungal plant disease that affects cereal crops such as wheat, barley, and oats. The fungus causes *Tapesia* to allude and can substantially reduce crop yield and quality. The disease is named for its distinctive symptoms on cereal

grains: tiny, dark, indented patches with a white dot in the center approximating a bird's eye. Commonly found on the kernels of cereal crops, these patches can render them unfit for human consumption or use as seed. Utilizing fungicides at the correct growth stages, crop rotation, and sowing resistant varieties are standard methods for controlling Bird's eye spots. Timely harvesting, appropriate storage, and seed treatment with fungicides can help minimize the disease's impact on crop yields and grain quality, ensuring the economic viability of cereal production[3].

Grey blight, a fungal disease caused by the fungus *Pestalotiopsis* spp, is a significant concern in tea cultivation regions because it affects tea plants. The condition causes grey-to-brown lesions or blotches on leaves, stems, buds, and juvenile branches, which can become necrotic over time. The illness thrives in warm, humid environments and is spread by wind-borne spores or water droplets. Proper plant spacing, good air circulation, and avoiding overfertilization are all preventative methods. Integrated pest management (IPM) practices are required to prevent fungicide resistance and environmental consequences. Pruning and removing diseased plant debris may also help with disease management. If uncontrolled, grey blight may lower tea output and quality, a significant worry for tea growers. Monitoring symptoms and implementing appropriate management measures are critical to reducing the effect of this fungal disease on tea crops[4].

Anthrachnose is a widespread fungal disease that affects various plant types, including trees, shrubs, vegetables, and fruits. Numerous species of *Colletotrichum* fungi cause it. On leaves, stems, flowers, and fruit, Anthrachnose typically manifests as lesions that are dark and depressed. Depending on the specific host plant and environmental conditions, the severity of the disease can vary[5].

Brown blight is a fungal disease that affects numerous plant species, including trees, vegetation, and ornamental plants. Numerous species of fungi cause it, and its symptoms typically involve the appearance of brown to black lesions on leaves, stems, and other plant parts[6].

II. LITERATURE REVIEW

S. Mukhopadhyay [7] Used image processing technology, a novel method for the automated detection of disorders in tea leaves was presented and combined the use of the NSGA-II (Nondominated Sorting Genetic Algorithm) for image clustering to identify disease-affected regions on tea leaves. Then, PCA (Principal Component Analysis) was used to reduce the number of features, and a multi-class SVM (Support Vector Machine) was applied for accurate disease identification. The results of this study determine that the algorithm was capable of accurately identifying the type of diseases present in tea leaves, with an accuracy of 83%. Five distinct tea leaf maladies, including Thrips, Heliopolis, Red Rust, Red Spider, and Sunlight Scorching, were investigated in this study.

Z. Xue [8] Introduced the YOLO-Tea model, an improved version of You Only Look Once version 5, which has been enhanced by integrating Self-attention and convolution as fundamental components in deep learning models and receptive field block modules. A global context network was incorporated into the model to decrease resource utilization. Compared to YOLOv5s, the proposed YOLO-Tea improved by 0.3% to 15.0% across all test data,

outperforming Faster R-CNN and SSD by 5.5%, 1.8%, and 7.0%, and 7.7%, 7.8%, and 5.2%, respectively.

W. Bao [9] Introduced the X-module's feature fusion module, which generates rich feature maps using Multiple fusions of multiscale characteristics. Each feature map channel is assigned an adaptively optimized weight by the attention module, reducing redundant features. Data augmentation methods are used to address insufficient samples. Experimental results show that AX-RetinaNet outperforms existing target detection and identification networks for tea leaf diseases in natural scene images. The F1-score is 0.954, and the mAP value is 93.83%.

G. Hu [10] Introduces a deep learning method for detecting and analyzing TLB leaves using a Retinex algorithm and Faster Region-based CNN. The technique enhances the accuracy of detecting diseased leaves that are obscured, tiny, or partially covered. Additionally, the procedure assists in severity classification and disease severity analysis. Compared to conventional machine learning methods, experimental results indicate a 6% improvement in detection accuracy and a 9% improvement in severity grading accuracy.

G. Hu [11] Used a depthwise separable convolution module and a multiscale feature extraction module, and a deep CNN method for tea leaf disease identification is presented. The technique obtains an average identification accuracy of 92.5%, surpassing both conventional machine learning methods and traditional deep learning approaches, demonstrating its potential in tea leaf disease detection.

Y. Wang [12] Proposed Tea pests and illnesses are detected using an integrated learning-based pest detection technique, concentrating on Leaf blight and *Apolygus lucorum*. The YOLOv5 unsupervised or weakly supervised model was utilized, and the GAM and CBAM attention mechanisms were introduced to enhance the detection of *Apolygus lucorum* and Leaf blight. The WBF (weighted box fusion) algorithm was utilized to process prediction results. The integrated model improves detection with an accuracy of 79.3%, outperforming individual models by 8.7% and 9.3%, respectively. This method facilitates tea disease diagnosis and enhances feature extraction capacities.

W. Bao[13] presents a TLB detection and monitoring using a UAV remote sensing approach based on DDMA-YOLO, thereby reducing labor and time usage. The RCAN method will be used to rebuild high-resolution tea pictures, with Retinex for image contrast and irregular illumination. The DDMA-YOLO model enhances accuracy by adding a dual-dimensional mixed attention (DDMA) module with a multiscale RFB module to the Neck.

The study[14] employs a neural network technique to detect diseases and parasites in tea leaves, evaluating different feature extraction networks and detection architectures. The system accurately distinguishes health conditions in complex backgrounds and with occlusion, with YOLO-based detection algorithms achieving a mAP of 86% and an IOU of 50% in real-time.

In the study[15], a different deep CNN model named LeafNet was devised to identify diverse tea plant disease varieties from leaf images accurately. The model integrates distinctive DSIFT (Dense Scale-Invariant Feature Transform) features and leverages a bag of visual word

models for efficient disease classification. LeafNet uses both SVM and MLP classifiers to accomplish its classification task. The LeafNet algorithm has an average classification accuracy of 90.16 percent. Demonstrates a remarkable level of accuracy. This outstanding performance transcends that of the MLP and SVM algorithms, establishing LeafNet as a promising and potentially invaluable tool for improving the efficacy of disease diagnosis in tea plants.

III. METHODOLOGY

This part covers the rigorous methodology used in the study in great detail. The dataset utilized to train our model is given, and the steps taken to prepare it for analysis are also explained.

A. Dataset

A meticulously curated dataset was employed during the research, comprising approximately 900 high-resolution images distributed across eight distinct categories. These categories encompass seven classes representing various tea leaf illnesses and one category dedicated to healthy tea leaves, a pivotal component for the training and evaluation of the machine learning model. Within this dataset, each class boasts more than 100 meticulously captured and annotated images. This rich and diverse dataset is the foundation upon which our model's ability to discern and classify tea leaf conditions is honed and fine-tuned. The dataset was meticulously assembled and harvested from the tea plantation located at Johnstone Boiyon Farm, nestled in the picturesque locale of Koiwa within Bomet County. The tea leaves featured in the dataset originate from a clone denoted as 1510, chosen for its representativeness of the tea plants commonly found in the region. The meticulous selection of this specific clone and the data collection location ensures that the dataset accurately mirrors real-world conditions and offers a reliable and comprehensive basis for training and validating the model's disease detection capabilities. With its robust collection of images spanning various tea leaf conditions, this dataset contributes to advancing tea plant disease detection. It holds significant promise for future research and applications in agricultural technology.

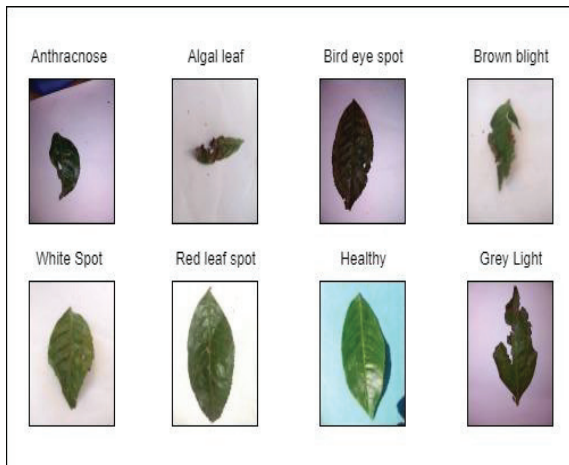


Fig. 1. Example of the dataset

B. Preprocessing

Practical data assessment is a critical foundation for successfully identifying tea leaf illnesses, and it hinges significantly on preprocessing. In the context of tea leaf disease detection, preprocessing encompasses a series of

essential steps designed to enhance the accuracy and reliability of the data before analysis. These preprocessing procedures are pivotal in ensuring that the data is in an optimal state for subsequent analysis, thereby exerting a substantial impact on the precision and efficiency of the diagnostic process. One fundamental aspect of data enhancement in machine learning and deep learning is data augmentation. This technique plays a pivotal role in expanding and diversifying the training dataset without the need for additional real-world data. Working within the constraints of limited data, data augmentation bolsters model performance fosters better generalization, and enhances resilience. This is achieved by introducing variations and perturbations into the dataset while preserving the original labels, thereby creating new and augmented training examples. Data augmentation encompasses a range of typical methods, including Random Rotation, Horizontal and Vertical Flipping, resizing, Shearing, Random Zoom, Color Transformations, and Noise Addition. Implementing these data augmentation techniques enriches the dataset with a broader spectrum of training examples, enabling the model to learn and adapt to a more comprehensive range of scenarios. This enhances the model's performance and contributes to its ability to generalize well to unseen data and exhibit robustness in the face of varying conditions, ultimately improving the accuracy and efficacy of tea leaf disease diagnosis.

C. Optimizer

It is wise to use Adaptive Moment Estimation (ADAM) as the model's optimizer. ADAM is widely used as an efficient optimization tool in deep neural networks. It is renowned for producing powerful results effectively. ADAM excels in circumstances with large datasets because of its capacity to efficiently modify the learning rates for different model parameters. The mean of gradients and the mean of squared gradients are two crucial statistics that must be calculated as part of the optimization process in ADAM. These data are essential for fine-tuning the model during training and assisting in the convergence of the model's weights. These calculations are typically depicted using equations such as Eq. (1) and Eq. (2).

$$m_t = \beta_1 \times m_{t-1} + g_t \times (1 - \beta_1) \quad (1)$$

$$v_t = \beta_2 \times v_{t-1} + g_t^2 \times (1 - \beta_2) \quad (2)$$

Where i_t refers to the first-moment estimate at iteration t , v_t represents the second-moment estimate at iteration t , and i_t signifies the gradient at iteration t . Parameters β_1 and β_2 are the exponential decay rates used in the moving average calculations. First and second moments (m_t and v_t) are estimated after initializations at $t=0$ and then subjected to bias correction. The bias-corrected first-moment estimate ($\hat{m}_t$) and second-moment estimate ($\hat{v}_t$) are obtained from Eq. (3) and (4), respectively. Applying this adjustment, particularly in the first optimization stages, is crucial to guarantee reliable and objective estimations.

$$\hat{m}_t = m_t / (1 - \beta_1^t) \quad (3)$$

$$\hat{v}_t = v_t / (1 - \beta_2^t) \quad (4)$$

Finally, parameters are calculated using bias-corrected first and second-moment estimates, as shown in Eq. (5).

$$x_{t+1} = x_t - \frac{\eta}{\sqrt{v_t + \epsilon}} \times \widehat{m}_t \quad (5)$$

Where x_t is the parameter at iteration t , η is the learning rate, and ϵ is a small constant added to avoid division by 0[16].

IV. MODEL

This system used a Conv2D-based model to detect the diseases of tea leaves. The use of a Conv2D model, or Convolutional 2D model, is an essential part of the process. Conv2D is a particular kind of Convolutional Neural Network (CNN) architecture that excels at handling 2D input data, such as photographs. This decision makes sense given that a large portion of tea leaf disease diagnosis requires dealing with tea leaf photos. By advancing a convolution kernel across the two-dimensional grid of the image, the Conv2D model processes tea leaf pictures. It calculates the dot product between the seed and the area of the picture that overlaps at each place, which is an efficient way to identify regional patterns and features. The result is a feature map that emphasizes the critical information for illness diagnosis when this procedure is repeated throughout the whole picture. This study's usage of a 17-layer CNN model(Fig.2) highlights the significance of deep learning architectures in effectively addressing the difficulty of tea leaf disease detection.

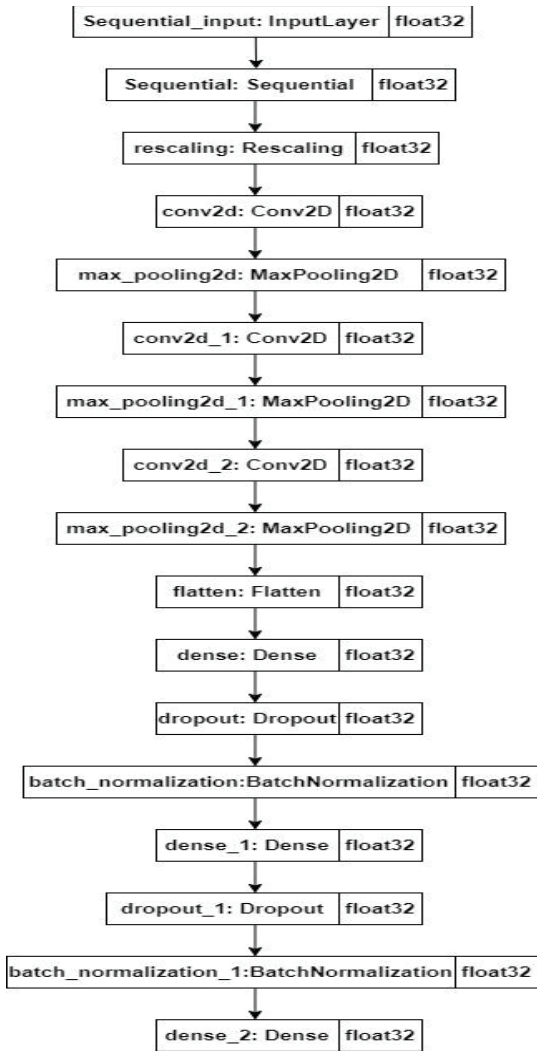


Fig. 2. CNN model

V. RESULTS

To correctly categorize tea leaf illnesses, the machine learning model employed in this study for disease detection on tea leaves underwent a thorough training procedure. There were a lot of essential factors in the training process, but the one that stood out was the 320-epoch limit on the number of training epochs. During each epoch, the model learned from the training data and made predictions, then compared to the actual labels or ground truth values. Any discrepancies between the predicted and actual values were used to update the model's parameters through backpropagation, wherein the model learns to reduce prediction errors. This thorough training procedure yielded a phenomenal accuracy percentage for the model of 84.5%(Fig.3, Fig.4). The percentage of instances of tea leaf disease identified correctly out of all the cases is known as accuracy in this context. An accuracy rating of 84.5% means that, on average, the model accurately predicted the outcomes for 84.5% of the dataset's samples of tea leaf disease. As shown in Table I, this high degree of precision demonstrates the model's capability to discriminate between various tea leaf illnesses, highlighting its potential value as a valuable tool for tea plant disease diagnostics.

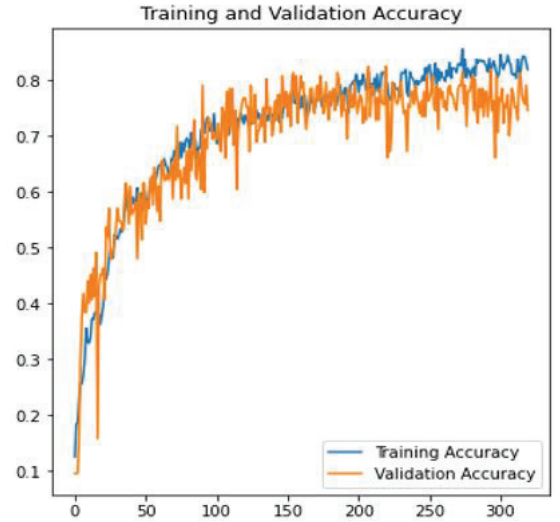


Fig. 3. Accuracy Graph

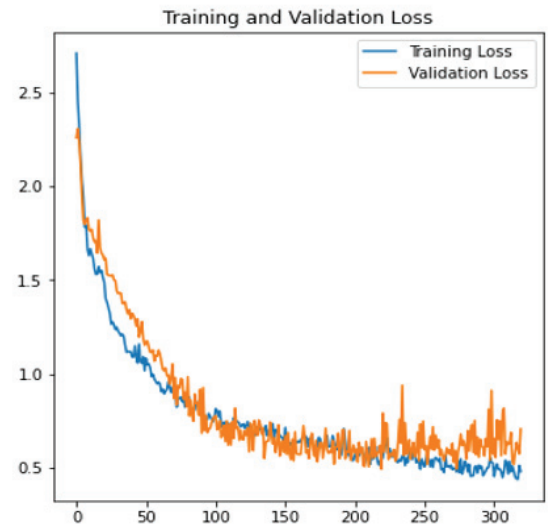


Fig. 4. Loss Graph

TABLE I. PRECISION, RECALL, F1-SCORE TABLE

Classes	Precision	Recall	F1-Score
Anthrachnose	67	44	53
Algal leaf spot	71	92	80
Bird's eyespot	80	36	50
Brown blight	84	70	76
Gray blight	58	95	72
Healthy	100	100	100
Red leaf spot	100	93	97
White spot	71	71	71

VI. CONCLUSION AND FUTURE SCOPE

This study article explores the crucial field of tea leaf disease detection via the EfficientNet architecture, known for its effectiveness and efficiency. Diseases affecting tea leaves provide a substantial risk to the tea business, negatively affecting the quantity and quality of tea crops. To tackle this particular difficulty, an examination has been conducted on the capabilities of EfficientNet, a cutting-edge deep-learning architecture renowned for its exceptional performance in tasks related to picture categorization. The EfficientNet architecture was prudent, as it demonstrated outstanding performance in capturing detailed characteristics and patterns within the tea leaf pictures, even while operating under constrained computing resources. The model shows a notable capacity to differentiate between several illness categories with a significant level of precision. This is critical for the timely and precise diagnosis of tea leaf illnesses. It is possible to investigate further modifications and optimizations of the model to enhance its resilience and versatility in various tea plantation settings.

Furthermore, the incorporation of real-time monitoring and data-collecting technology has the potential to initiate a new phase of proactive disease control within tea farms. The tea leaf disease detection system, leveraging advanced technologies like deep learning and computer vision, offers promising prospects for the tea industry's future. Here are some critical areas of future scope and development:

- **Enhanced Disease Classification:** Disease categorization may be expanded in future studies. Many systems only detect a few illnesses. Farmers will get more complete remedies by expanding the model to identify new tea plant illnesses and abnormalities [17].
- **Real-Time Monitoring:** Using drones and IoT devices to monitor tea plants in real time may transform the game. These sensors can monitor environmental conditions, disease development, and insect infestations for preventive management [18].
- **Mobile Applications:** Farmers may benefit from user-friendly smartphone apps that swiftly analyse tea leaf photos for illnesses. Applications may give quick disease management feedback and suggestions [19].

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